

ECO387C - Fall 2024

Discussion 3

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Two topics are going to be covered in this discussion:

- Log-linearization
- The time-averaging problem by Working (1960)

1 Log-linearization

1.1 Preliminaries

One of the underlying ideas in macroeconomics is that the time series data we observe such as GDP, consumption, and price levels are growing with a trend and potentially experiencing a cycle around the trend. Here we will take the existence and knowledge of the trend as given. Define the following quantities (in levels):

- Original series x_t
- Trend $\bar{x}_t$
- Cycle $\Delta x_t := x_t - \bar{x}_t$
- Percentage deviation from trend $\hat{x}_t := \frac{x_t - \bar{x}_t}{\bar{x}_t} = \frac{\Delta x_t}{\bar{x}_t}$

The relationship between these is

$$x_t = \bar{x}_t \left(1 + \frac{x_t - \bar{x}_t}{\bar{x}_t} \right) = \bar{x}_t (1 + \hat{x}_t)$$

For small deviations Δx_t from trend, we can approximate:

$$x_t = \bar{x}_t e^{\frac{\Delta x_t}{\bar{x}_t}} \approx \bar{x}_t e^{\hat{x}_t}$$

Also, note that the log-difference is roughly equal to the growth rate:

$$\tilde{x}_t = \frac{x_t - \bar{x}_t}{\bar{x}_t} \approx \ln \left(\frac{x_t}{\bar{x}_t} \right) = \ln(x_t) - \ln(\bar{x}_t)$$

1.2 Motivation

Often the equilibrium of a model is described by a nonlinear system of equations. Such systems are hard to solve analytically, so we approximate these equations with their linear counterpart around the steady state. This allows us to understand the behavior of the economy "close" to the steady state.

I will explain different methods of log-linearization using a simple example, the Cobb-Douglas production function:

$$Y_t = A_t K_t^\alpha N_t^{1-\alpha}$$

1.3 Method 1

1. Replace all time-dependent variables in the equation with their approximation $x_t = \bar{x}_t e^{\hat{x}_t}$. This gives:

$$\bar{Y}_t e^{\hat{Y}_t} = \bar{A}_t e^{\hat{A}_t} \left(\bar{K}_t e^{\hat{K}_t} \right)^\alpha \left(\bar{N}_t e^{\hat{N}_t} \right)^{1-\alpha}$$

2. Cancel the trend by dividing both sides by the steady-state values:

$$e^{\hat{Y}_t} = e^{\hat{A}_t} e^{\alpha \hat{K}_t} e^{(1-\alpha) \hat{N}_t}$$

3. Take logs of both sides to linearize the expression:

$$\hat{Y}_t = \hat{A}_t + \alpha \hat{K}_t + (1-\alpha) \hat{N}_t$$

1.4 Method 2

1. Take the first-order Taylor expansion around the trend. Start with the Cobb-Douglas production function:

$$Y_t = A_t K_t^\alpha N_t^{1-\alpha}$$

Expand Y_t around $\bar{Y}$, A_t around $\bar{A}$, K_t around $\bar{K}$, and N_t around $\bar{N}$:

$$Y_t \approx \underbrace{\bar{A} \bar{K}^\alpha \bar{N}^{1-\alpha}}_{=\bar{Y}} + \frac{\partial Y}{\partial A} \Big|_{\bar{A}, \bar{K}, \bar{N}} (A_t - \bar{A}) + \frac{\partial Y}{\partial K} \Big|_{\bar{A}, \bar{K}, \bar{N}} (K_t - \bar{K}) + \frac{\partial Y}{\partial N} \Big|_{\bar{A}, \bar{K}, \bar{N}} (N_t - \bar{N})$$

Calculating the partial derivatives and substituting these into the Taylor expansion:

$$Y_t \approx \bar{Y} + \frac{\bar{Y}}{\bar{A}} (A_t - \bar{A}) + \alpha \frac{\bar{Y}}{\bar{K}} (K_t - \bar{K}) + (1-\alpha) \frac{\bar{Y}}{\bar{N}} (N_t - \bar{N})$$

2. Cancel the trend component:

$$\frac{Y_t - \bar{Y}}{\bar{Y}} \approx \frac{(A_t - \bar{A})}{\bar{A}} + \alpha \frac{(K_t - \bar{K})}{\bar{K}} + (1-\alpha) \frac{(N_t - \bar{N})}{\bar{N}}$$

Using the hat notation:

$$\hat{Y}_t \approx \hat{A}_t + \alpha \hat{K}_t + (1-\alpha) \hat{N}_t$$

1.5 Method 3

1. Take logs:

$$\log(Y_t) = \log(A_t) + \alpha \log(K_t) + (1 - \alpha) \log(N_t)$$

2. Apply total differentiation to both sides:

$$\frac{dY_t}{Y_t} = \frac{dA_t}{A_t} + \alpha \frac{dK_t}{K_t} + (1 - \alpha) \frac{dN_t}{N_t}$$

3. Evaluate the resulting equation at the steady state. Using $\hat{x}_t := \frac{x_t - \bar{x}_t}{\bar{x}_t} = \frac{dx_t}{\bar{x}_t}$

$$\hat{Y}_t = \hat{A}_t + \alpha \hat{K}_t + (1 - \alpha) \hat{N}_t$$

2 Time-averaging

This problem was identified by Working (1960). He realized that the use of averages in data construction can introduce correlations not present in the original series. For example, the measured data x_{t+1} on actual consumption c_t were constructed as

$$x_{t+1} = \frac{c_{t+1} + c_t}{2}$$

and it is observed only every second period: $x_{t+1}, x_{t+3}, \dots$. If true consumption follows approximately a random walk:

$$c_t = c_{t-1} + \epsilon_t,$$

where ϵ_t is white noise, then this causes practical problems for the testing of consumption theories. Before we start, let us briefly state how white noise is formally defined.

Definition 1 *In the scalar case, white noise ϵ_t satisfies three properties:*

- *Its mean exists and is zero: $E[\epsilon_t] = 0$,*
- *Its variance exists and is constant: $V[\epsilon_t] = E[\epsilon_t^2] = \sigma^2$,*
- *It is uncorrelated over time: $Cov[\epsilon_t, \epsilon_s] = E[\epsilon_t \epsilon_s] = 0$ for all $t \neq s$.*

Suppose now that true consumption follows a random walk as in the above equation, but we erroneously believe that the measured data x_t followed a random walk:

$$x_{t+1} = x_t + u_{t+1}$$

To test this hypothesis, we compute the first-order autocovariance of u_{t+1} , $\text{Cov}[u_{t+1}, u_{t-1}]$. If x_{t+1} followed a random walk, this expression should be white noise, and therefore, the first-order autocovariance should be zero. However, it turns out that this is not true, and the problem arises from time averaging. Consider first what u_{t+1} actually is:

$$\begin{aligned}
u_{t+1} &= x_{t+1} - x_{t-1} \\
&= \frac{c_{t+1} + c_t}{2} - \frac{c_{t-1} + c_{t-2}}{2} \\
&= \frac{1}{2}[c_{t+1} + c_t - c_{t-1} - c_{t-2}] \\
&= \frac{1}{2}[2c_t + \epsilon_{t+1} - c_{t-1} - c_{t-2}] \\
&= \frac{1}{2}[c_{t-1} + 2\epsilon_t + \epsilon_{t+1} - c_{t-2}] \\
&= \frac{1}{2}[c_{t-2} + \epsilon_{t-1} + 2\epsilon_t + \epsilon_{t+1} - c_{t-2}] \\
&= \frac{1}{2}[\epsilon_{t+1} + 2\epsilon_t + \epsilon_{t-1}].
\end{aligned}$$

This expression already gives an important insight: the error of the series x_{t+1} consists of the true error and two of its past lags. It is now almost obvious that $\text{Cov}[u_{t+1}, u_{t-1}]$ cannot be zero. Let's prove this formally:

$$\begin{aligned}
\text{Cov}[u_{t+1}, u_{t-1}] &= E[u_{t+1}u_{t-1}] \\
&= E\left[\frac{1}{2}[\epsilon_{t+1} + 2\epsilon_t + \epsilon_{t-1}] \cdot \frac{1}{2}[\epsilon_{t-1} + 2\epsilon_{t-2} + \epsilon_{t-3}]\right] \\
&= \frac{1}{4}E[\epsilon_{t-1}^2] \\
&= \frac{1}{4}\sigma^2,
\end{aligned}$$

because all cross-products cancel out. As a consequence, x_t does not follow a random walk since u_t is not white noise. If we are not aware of the time averaging problem, we might erroneously reject the random walk hypothesis. Note that double lagging the data solves this problem. If we only work with the sequence $x_{t-1}, x_{t+3}, \dots$, the errors $u_{t-1}, u_{t+3}, \dots$ are uncorrelated. Confirming this is left as an exercise.

References

Holbrook Working. Note on the correlation of first differences of averages in a random chain. *Econometrica*, 28(4):916–918, 1960.